

FormFix-AI

Multi-Model Real-Time Gym Pose Detection and Correction

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UVCE Mini Project Presentation

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Problem Statement

- No trainer supervision during home workouts
- Wrong form increases injury risk
- Manual rep counting is distracting
- Existing apps mostly detect **what** exercise is happening
- FormFix AI focuses on **how well** the exercise is performed

Objectives and Paper Contributions Used

Project Objectives

- real-time exercise recognition
- accurate rep counting
- posture correction
- voice and visual feedback
- progress tracking in SQL

Implemented from Papers

- MediaPipe/OpenPose landmarks
- LSTM/BiLSTM/Transformer sequence learning
- joint-angle posture rules
- real-time corrective feedback

Processed Dataset

- 1140 total sequences
- sequence shape: (1140, 30, 132)
- 6 exercise classes

Class Counts

- barbell_squat: 152
- bicep_curl: 225
- hammer_curl: 78
- lateral_raise: 170
- pull_up: 212
- push_up: 303

Preprocessing Steps

- 1 collect videos
- 2 detect pose landmarks
- 3 compute frame features
- 4 build 30-frame windows
- 5 save exercise_sequences.npz

System Architecture



Why Three Models?

Model	Role	Why used
LSTM	Baseline	Fast and simple temporal classifier
BiLSTM	Main model	Learns motion in forward and backward direction
Transformer	Advanced model	Captures long-range dependencies using attention

Live Modes: Fast, Best, Hybrid, Ensemble

Model Architectures



LSTM

Baseline for speed and standard temporal learning



BiLSTM

Primary model for stronger motion context



Transformer

Attention-based comparison with highest current accuracy

Rep Counting and Posture Correction Logic

Rep Counting

- tracked angle per exercise
- smoothing
- up/down stage transitions
- threshold validation

Posture Correction

- elbow = shoulder–elbow–wrist
- knee = hip–knee–ankle
- hip = shoulder–hip–knee
- shoulder = elbow–shoulder–hip

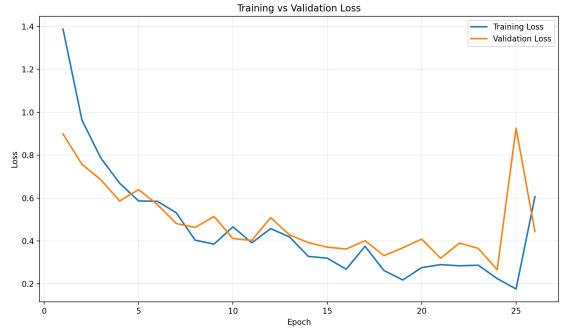
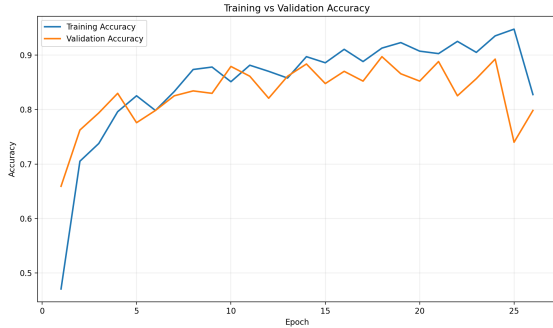
$$\theta = \cos^{-1} \left(\frac{(A - B) \cdot (C - B)}{\|A - B\| \|C - B\|} \right)$$

Training Snapshots

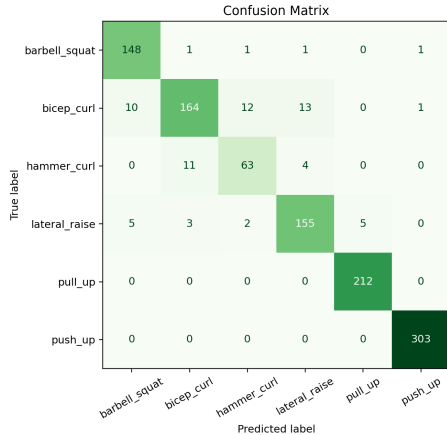
```
Epoch 7/22
187/187 [=====] - 8s 28ms/step - loss: 1.8486 - accuracy: 0.6286 - val_loss: 0.8880 - val_accuracy: 0.6622 - lr: 0.0010
Epoch 8/22
187/187 [=====] - 4s 28ms/step - loss: 0.8195 - accuracy: 0.7390 - val_loss: 0.7413 - val_accuracy: 0.7130 - lr: 0.0010
Epoch 9/22
187/187 [=====] - 4s 19ms/step - loss: 0.4087 - accuracy: 0.7722 - val_loss: 0.4806 - val_accuracy: 0.8157 - lr: 0.0010
Epoch 10/22
187/187 [=====] - 4s 19ms/step - loss: 0.3934 - accuracy: 0.8184 - val_loss: 0.6236 - val_accuracy: 0.8277 - lr: 0.0010
Epoch 11/22
187/187 [=====] - 4s 28ms/step - loss: 0.5868 - accuracy: 0.8364 - val_loss: 0.5967 - val_accuracy: 0.8485 - lr: 0.0010
Epoch 12/22
187/187 [=====] - 4s 19ms/step - loss: 0.4726 - accuracy: 0.8486 - val_loss: 0.2987 - val_accuracy: 0.8914 - lr: 0.0010
Epoch 13/22
187/187 [=====] - 4s 20ms/step - loss: 0.4130 - accuracy: 0.8667 - val_loss: 0.3230 - val_accuracy: 0.8881 - lr: 0.0010
Epoch 14/22
187/187 [=====] - 4s 20ms/step - loss: 0.3979 - accuracy: 0.8689 - val_loss: 0.2603 - val_accuracy: 0.9076 - lr: 0.0010
Epoch 15/22
187/187 [=====] - 4s 21ms/step - loss: 0.3691 - accuracy: 0.8848 - val_loss: 0.2862 - val_accuracy: 0.8914 - lr: 0.0010
Epoch 16/22
187/187 [=====] - 4s 20ms/step - loss: 0.3684 - accuracy: 0.8984 - val_loss: 0.2482 - val_accuracy: 0.9082 - lr: 0.0010
Epoch 17/22
187/187 [=====] - 4s 20ms/step - loss: 0.3809 - accuracy: 0.9016 - val_loss: 0.2510 - val_accuracy: 0.8920 - lr: 0.0010
Epoch 18/22
187/187 [=====] - 4s 19ms/step - loss: 0.3866 - accuracy: 0.8983 - val_loss: 0.2409 - val_accuracy: 0.9075 - lr: 0.0010
Epoch 19/22
186/187 [=====] - ETA: 0s - loss: 0.3287 - accuracy: 0.8987
Epoch 19: ReduceLROnPlateau reducing learning rate to 0.00050000000000000005
187/187 [=====] - 4s 19ms/step - loss: 0.3282 - accuracy: 0.8989 - val_loss: 0.2587 - val_accuracy: 0.9048 - lr: 0.0010
Epoch 20/22
187/187 [=====] - 4s 28ms/step - loss: 0.2174 - accuracy: 0.9279 - val_loss: 0.1766 - val_accuracy: 0.9236 - lr: 5.0000e-04
Epoch 21/22
187/187 [=====] - 4s 19ms/step - loss: 0.1082 - accuracy: 0.9351 - val_loss: 0.1508 - val_accuracy: 0.9430 - lr: 5.0000e-04
Epoch 22/22
187/187 [=====] - 4s 20ms/step - loss: 0.1077 - accuracy: 0.9403 - val_loss: 0.1279 - val_accuracy: 0.9511 - lr: 5.0000e-04
Epoch 23/22
187/187 [=====] - 4s 19ms/step - loss: 0.1029 - accuracy: 0.9418 - val_loss: 0.1472 - val_accuracy: 0.9424 - lr: 5.0000e-04
Epoch 24/22
187/187 [=====] - 4s 19ms/step - loss: 0.1053 - accuracy: 0.9495 - val_loss: 0.1248 - val_accuracy: 0.9530 - lr: 5.0000e-04
Epoch 25/22
187/187 [=====] - 4s 19ms/step - loss: 0.1012 - accuracy: 0.9506 - val_loss: 0.1172 - val_accuracy: 0.9564 - lr: 5.0000e-04
Epoch 26/22
187/187 [=====] - 4s 28ms/step - loss: 0.1429 - accuracy: 0.9531 - val_loss: 0.1397 - val_accuracy: 0.9491 - lr: 5.0000e-04
Epoch 27/22
187/187 [=====] - 4s 28ms/step - loss: 0.1456 - accuracy: 0.9499 - val_loss: 0.1211 - val_accuracy: 0.9611 - lr: 5.0000e-04
Epoch 28/22
185/187 [=====] - ETA: 0s - loss: 0.1488 - accuracy: 0.9456
Epoch 28: ReduceLROnPlateau reducing learning rate to 0.00025000000000000002
187/187 [=====] - 4s 20ms/step - loss: 0.1677 - accuracy: 0.9454 - val_loss: 0.1199 - val_accuracy: 0.9530 - lr: 5.0000e-04
2026-04-13 00:14:14.317 | train_lstm | INFO | Saved BiLSTM model to models/exercise_bi_lstm.keras
2026-04-13 00:14:14.317 | train_lstm | INFO | Saved LSTM model to models/exercise_lstm.keras
2026-04-13 00:14:14.320 | train_lstm | INFO | Final validation accuracy: 0.9799
```

```
Epoch 1/20
2026-04-13 00:14:44.4823117 | tensorflow/core/grappler/optimizers/custom_graph_optimizer_registry.cc:1117 | Plugin optimizer for device_type GPU is enabled.
187/187 [=====] - 8s 36ms/step - loss: 0.8476 - accuracy: 0.6743 - val_loss: 0.5449 - val_accuracy: 0.7836
Epoch 2/20
187/187 [=====] - 4s 30ms/step - loss: 0.4480 - accuracy: 0.8364 - val_loss: 0.3719 - val_accuracy: 0.8372
Epoch 3/20
187/187 [=====] - 4s 30ms/step - loss: 0.3516 - accuracy: 0.8697 - val_loss: 0.3181 - val_accuracy: 0.8721
Epoch 4/20
187/187 [=====] - 8s 41ms/step - loss: 0.2845 - accuracy: 0.8964 - val_loss: 0.3335 - val_accuracy: 0.8854
Epoch 5/20
187/187 [=====] - 13s 71ms/step - loss: 0.2680 - accuracy: 0.9084 - val_loss: 0.2444 - val_accuracy: 0.9109
Epoch 6/20
187/187 [=====] - 7s 36ms/step - loss: 0.2148 - accuracy: 0.9179 - val_loss: 0.1676 - val_accuracy: 0.9358
Epoch 7/20
187/187 [=====] - 7s 39ms/step - loss: 0.1892 - accuracy: 0.9298 - val_loss: 0.1714 - val_accuracy: 0.9471
Epoch 8/20
187/187 [=====] - 7s 37ms/step - loss: 0.1999 - accuracy: 0.9211 - val_loss: 0.2646 - val_accuracy: 0.9048
Epoch 9/20
187/187 [=====] - 7s 36ms/step - loss: 0.1762 - accuracy: 0.9316 - val_loss: 0.1584 - val_accuracy: 0.9438
Epoch 10/20
187/187 [=====] - 8s 45ms/step - loss: 0.1610 - accuracy: 0.9273 - val_loss: 0.1876 - val_accuracy: 0.9438
Epoch 11/20
187/187 [=====] - 7s 37ms/step - loss: 0.1608 - accuracy: 0.9383 - val_loss: 0.1861 - val_accuracy: 0.9383
Epoch 12/20
187/187 [=====] - 7s 39ms/step - loss: 0.1458 - accuracy: 0.9468 - val_loss: 0.1884 - val_accuracy: 0.9417
Epoch 13/20
187/187 [=====] - 7s 36ms/step - loss: 0.1187 - accuracy: 0.9641 - val_loss: 0.1449 - val_accuracy: 0.9558
Epoch 14/20
187/187 [=====] - 4s 33ms/step - loss: 0.1087 - accuracy: 0.9588 - val_loss: 0.1647 - val_accuracy: 0.9497
Epoch 15/20
187/187 [=====] - 7s 35ms/step - loss: 0.1328 - accuracy: 0.9460 - val_loss: 0.1492 - val_accuracy: 0.9497
Epoch 16/20
187/187 [=====] - 4s 34ms/step - loss: 0.1117 - accuracy: 0.9568 - val_loss: 0.1562 - val_accuracy: 0.9497
Epoch 17/20
187/187 [=====] - 7s 36ms/step - loss: 0.1064 - accuracy: 0.9688 - val_loss: 0.1378 - val_accuracy: 0.9578
Epoch 18/20
187/187 [=====] - 4s 33ms/step - loss: 0.0923 - accuracy: 0.9648 - val_loss: 0.1248 - val_accuracy: 0.9648
Epoch 19/20
187/187 [=====] - 7s 36ms/step - loss: 0.0782 - accuracy: 0.9681 - val_loss: 0.1489 - val_accuracy: 0.9491
Epoch 20/20
187/187 [=====] - 4s 35ms/step - loss: 0.1102 - accuracy: 0.9568 - val_loss: 0.1598 - val_accuracy: 0.9551
Epoch 21/20
187/187 [=====] - 8s 44ms/step - loss: 0.0912 - accuracy: 0.9661 - val_loss: 0.1786 - val_accuracy: 0.9418
Epoch 22/20
187/187 [=====] - 4s 32ms/step - loss: 0.0908 - accuracy: 0.9661 - val_loss: 0.1256 - val_accuracy: 0.9658
Epoch 23/20
187/187 [=====] - 4s 34ms/step - loss: 0.0817 - accuracy: 0.9698 - val_loss: 0.1816 - val_accuracy: 0.9725
Epoch 24/20
187/187 [=====] - 4s 33ms/step - loss: 0.0782 - accuracy: 0.9732 - val_loss: 0.1421 - val_accuracy: 0.9611
Epoch 25/20
187/187 [=====] - 7s 39ms/step - loss: 0.0691 - accuracy: 0.9779 - val_loss: 0.1351 - val_accuracy: 0.9638
Epoch 26/20
187/187 [=====] - 18s 52ms/step - loss: 0.0909 - accuracy: 0.9626 - val_loss: 0.1683 - val_accuracy: 0.9524
Epoch 27/20
187/187 [=====] - 7s 48ms/step - loss: 0.0898 - accuracy: 0.9676 - val_loss: 0.1596 - val_accuracy: 0.9538
Epoch 28/20
187/187 [=====] - 9s 68ms/step - loss: 0.1029 - accuracy: 0.9614 - val_loss: 0.1368 - val_accuracy: 0.9438
Epoch 29/20
187/187 [=====] - 4s 34ms/step - loss: 0.1128 - accuracy: 0.9678 - val_loss: 0.1383 - val_accuracy: 0.9564
Epoch 30/20
187/187 [=====] - 4s 32ms/step - loss: 0.0808 - accuracy: 0.9787 - val_loss: 0.1186 - val_accuracy: 0.9685
2026-04-13 00:18:18.804 | train_transformer | INFO | Saved transformer model to models/posture_transformer.keras
2026-04-13 00:18:18.804 | train_transformer | INFO | Final validation accuracy: 0.9688
```

Training Curves



Confusion Matrix and Current Metrics



Current Metrics

- LSTM Top-1: **84.30%**
- BiLSTM Top-1: **93.72%**
- BiLSTM Top-3: **99.82%**
- Transformer Top-1: **97.89%**
- Weighted F1: **93.64%**
- Macro F1: **91.69%**

How Accuracy Is Calculated

Top-1 Accuracy

$$\text{Top-1} = \frac{\text{Correct highest-confidence predictions}}{\text{Total predictions}} \times 100$$

Top-3 Accuracy

$$\text{Top-3} = \frac{\text{True class in top 3 predictions}}{\text{Total predictions}} \times 100$$

Precision / Recall / F1

$$\text{Precision} = \frac{TP}{TP + FP} \quad \text{Recall} = \frac{TP}{TP + FN} \quad F1 = 2 \cdot \frac{PR}{P + R}$$

Chapter1: Introduction

Chapter2: Literature Survey: Write about related papers or work in 1 paragraph per paper

Chapter3: Mathematical model/Architecture- System Requirements or Specification

Chapter4: Problem definition: Objective, Assumption

Chapter5: Algorithms and examples

Chapter6: Data structures used/Implementation

Chapter7: Simulation, Results, Performance Analysis, Graphs , Tables

Chapter8: Conclusions (3 Paragraph, 1 page)

Bibliography (IEEE format)

Author, Title, Publisher, page number, year of publication

Appendix

What the final system provides

- FastAPI backend
- React frontend
- SQL progress tracking
- recipe and meal section
- scan-based calorie logging
- end-of-exercise report

Improvement over papers

- not only model accuracy
- full real-time usable application

Conclusion and Future Work

Conclusion

- FormFix AI combines recognition, rep counting, posture correction, and reporting
- BiLSTM is the strongest balanced model for deployment
- Transformer shows highest current evaluation accuracy

Future Work

- add more exercises
- improve strict unseen benchmarking
- mobile deployment
- multi-view support
- user-specific calibration

Thank You

FormFix AI

Questions?